NetCore Calling: Number Provisioning and Dial Plans

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Summary

This article describes how to provision E.164 numbers in NetCore Calling 5.0, assign them to users and locations, configure location-based dial plans, set the optional outbound prefix "9", and maintain emergency addresses. It also includes verification commands, rollback steps, and guidance for resolving MTG-CALL-7002 Number Not Provisioned and MTG-CALL-7010 Emergency Address Missing.

Applies to: NetCore Calling 5.0, NetCore Control Hub, Enterprise voice administrators, Location-based dial plans, E.164 number provisioning

Overview

NetCore Calling 5.0 uses E.164 numbers for public number assignment and location-based dial plans for routing, emergency services validation, and user calling policy. Numbers can be assigned directly to users, shared services, or locations, depending on the calling design.

A valid configuration normally includes a provisioned E.164 number range, at least one calling location, a dial plan bound to that location, and an emergency address for each location. The outbound prefix "9" is optional and can be enabled per dial plan for sites that require legacy dialing behavior.

If a user attempts to place or receive calls with an unassigned or unavailable number, NetCore Calling may report MTG-CALL-7002 Number Not Provisioned. If a location or user lacks a valid emergency address, emergency calling activation and some number assignments may fail with MTG-CALL-7010 Emergency Address Missing.

- Use E.164 format for all provisioned numbers, for example +15551234001.
- Assign each user to one NetCore Calling location.
- Bind each dial plan to the correct location.
- Configure emergency address data before enabling users for outbound calling.
- For call forwarding, hunt groups, and voicemail after number provisioning, see KB-MTG-2005.
- For error-code details, including MTG-CALL-7002 Number Not Provisioned, MTG-CALL-7010 Emergency Address Missing, MTG-JOIN-1003 Meeting Locked or Host Lobby Active, MTG-JOIN-1007 Media Ports Blocked by Firewall, MTG-MED-4102 Media Path Degraded (high packet loss), MTG-MED-4105 No Audio Device Detected, MTG-SSO-6001 SAML Assertion Invalid or Expired, and MTG-SSO-6004 IdP Certificate Mismatch, see KB-MTG-2008.

Prerequisites

Before provisioning numbers, confirm that the organization has NetCore Calling 5.0 enabled in NetCore Control Hub and that the administrator account has Calling Administrator or Organization Administrator privileges.

Prepare the number inventory, site names, emergency addresses, user principal names, and the desired dialing behavior before running the configuration.

- NetCore Calling 5.0 tenant active in NetCore Control Hub.
- Administrator role: Calling Administrator or Organization Administrator.
- E.164 number inventory approved for the tenant.
- Location names defined, for example NYC-HQ or LON-BRANCH.
- Emergency address for every calling location.
- Decision on outbound prefix: disabled, or enabled with prefix "9".

- User accounts created before number assignment.

Example planning worksheet

Item	Example Value	Required
Location ID	NYC-HQ	Yes
Country/region	US	Yes
Emergency address	200 Hudson Ave, Floor 12, New York, NY 10013	Yes
Number range	+15551234000 through +15551234049	Yes
Main number	+15551234000	Recommended
Outbound prefix	9	Optional
Internal extension length	4	Recommended

Step-by-step Procedure

The following example provisions a New York location, imports a block of E.164 numbers, creates a location-based dial plan, enables the optional outbound prefix "9", and assigns numbers to users.

Run the commands from a workstation with the NetCore administrative CLI installed and authenticated to NetCore Control Hub.

- 1. Sign in to the tenant administration context.
- 2. Create the calling location.
- 3. Add the emergency address to the location.
- 4. Import the E.164 number block.
- 5. Assign the main number to the location.
- 6. Create the location-based dial plan.
- 7. Enable the optional outbound prefix "9" if required.
- 8. Assign numbers and extensions to users.
- 9. Commit and publish the calling configuration.

```
# 1. Sign in to NetCore Control Hub administration
ncctl login --tenant acme.example --role calling-admin

# 2. Create the calling location
ncctl calling location create \
  --location-id NYC-HQ \
  --display-name "New York Headquarters" \
  --country US \
  --timezone America/New_York

# 3. Add the emergency address
ncctl calling emergency-address set \
  --location-id NYC-HQ \
  --address-line1 "200 Hudson Ave" \
  --address-line2 "Floor 12" \
  --city "New York" \
  --state NY \
  --postal-code 10013 \
  --country US

# 4. Import E.164 numbers into the tenant inventory
ncctl calling numbers import \
  --range-start +15551234000 \
  --range-end +15551234049 \
  --location-id NYC-HQ \
  --type did

# 5. Assign a main number to the location
ncctl calling location update \
  --location-id NYC-HQ \
```

```
  --main-number +15551234000

# 6. Create a location-based dial plan
ncctl calling dial-plan create \
  --dial-plan-id DP-NYC-HQ \
  --location-id NYC-HQ \
  --country US \
  --extension-length 4 \
  --local-routing enabled

# 7. Enable optional outbound prefix "9"
ncctl calling dial-plan update \
  --dial-plan-id DP-NYC-HQ \
  --outbound-prefix 9 \
  --prefix-mode optional

# 8. Assign numbers and extensions to users
ncctl calling user assign-number \
  --user ana.patel@acme.example \
  --location-id NYC-HQ \
  --number +15551234011 \
  --extension 4011

ncctl calling user assign-number \
  --user marco.rossi@acme.example \
  --location-id NYC-HQ \
  --number +15551234012 \
  --extension 4012

# 9. Commit and publish the configuration
ncctl calling config validate --location-id NYC-HQ
ncctl calling config publish --location-id NYC-HQ --change-ticket CHG-2026-0418
```

Verification

After publishing the configuration, verify the location, emergency address, number inventory, user assignment, and dial-plan behavior. The following output shows a healthy configuration for location NYC-HQ.

If validation reports MTG-CALL-7002 Number Not Provisioned, confirm that the number exists in the tenant inventory and is not already assigned. If validation reports MTG-CALL-7010 Emergency Address Missing, configure or correct the emergency address on the affected location.

- Verify that the location status is Active.
- Verify that emergencyStatus is Valid.
- Verify that each user has an E.164 number and extension.
- Verify that outbound prefix behavior is optional when prefix "9" is enabled.
- Place test calls using both E.164 and extension dialing.

```
$ ncctl calling location show --location-id NYC-HQ
locationId: NYC-HQ
displayName: New York Headquarters
country: US
timezone: America/New_York
mainNumber: +15551234000
status: Active
emergencyStatus: Valid
publishedAt: 2026-04-18T14:22:31Z

$ ncctl calling numbers list --location-id NYC-HQ --status all
NUMBER         STATUS      ASSIGNED_TO                TYPE
+15551234000   Location    NYC-HQ                     did
+15551234001   Available   -                          did
+15551234002   Available   -                          did
+15551234011   Assigned    ana.patel@acme.example     did
+15551234012   Assigned    marco.rossi@acme.example   did

$ ncctl calling user show --user ana.patel@acme.example
user: ana.patel@acme.example
callingEnabled: true
locationId: NYC-HQ
```

```
number: +15551234011
extension: 4011
dialPlanId: DP-NYC-HQ
emergencyAddressInheritedFrom: NYC-HQ
status: Active

$ ncctl calling dial-plan show --dial-plan-id DP-NYC-HQ
dialPlanId: DP-NYC-HQ
locationId: NYC-HQ
country: US
extensionLength: 4
localRouting: enabled
outboundPrefix: 9
prefixMode: optional
status: Published

$ ncctl calling test-dial --from ana.patel@acme.example --to +15551234012
result: Allowed
normalizedDestination: +15551234012
route: on-net
policy: DP-NYC-HQ

$ ncctl calling test-dial --from ana.patel@acme.example --to 94012
result: Allowed
normalizedDestination: 4012
route: internal-extension
policy: DP-NYC-HQ
prefixStripped: true

$ ncctl calling config validate --location-id NYC-HQ
validationStatus: Passed
errors: 0
warnings: 0
```

Rollback

Use rollback when a number block, dial plan, or user assignment was published incorrectly. Roll back in the reverse order of provisioning: remove user assignments, remove or disable the dial plan, detach the main number, release imported numbers, and then delete the location if it is not needed.

Do not delete a location until all users, numbers, and calling services have been moved or unassigned. For hunt groups, call forwarding, and voicemail objects that may depend on these numbers, see KB-MTG-2005 before rollback.

- 1. Export the current configuration for audit.
- 2. Unassign affected user numbers.
- 3. Remove the dial plan or restore the previous dial plan.
- 4. Remove the location main number.
- 5. Release incorrectly imported numbers.
- 6. Delete the location only if no users or services remain assigned.
- 7. Validate and publish the rollback.

```
# 1. Export current configuration
ncctl calling config export \
  --location-id NYC-HQ \
  --output ./rollback-NYC-HQ-20260418.json

# 2. Unassign user numbers
ncctl calling user unassign-number --user ana.patel@acme.example
ncctl calling user unassign-number --user marco.rossi@acme.example

# 3. Disable or delete the dial plan
ncctl calling dial-plan update --dial-plan-id DP-NYC-HQ --status disabled
# Optional, only if the dial plan is no longer required:
# ncctl calling dial-plan delete --dial-plan-id DP-NYC-HQ

# 4. Remove the main number from the location
ncctl calling location update --location-id NYC-HQ --main-number none
```

```
# 5. Release number inventory from the location
ncctl calling numbers release \
  --range-start +15551234000 \
  --range-end +15551234049 \
  --location-id NYC-HQ

# 6. Delete the location only if unused
ncctl calling location delete --location-id NYC-HQ --force-if-empty

# 7. Validate and publish rollback
ncctl calling config validate --tenant acme.example
ncctl calling config publish --tenant acme.example --change-ticket CHG-2026-0418-RB
```

Known Limitations

NetCore Calling 5.0 requires numbers to be provisioned in E.164 format before assignment. Non-E.164 local formats, such as 5551234011, are not valid inventory numbers and can trigger MTG-CALL-7002 Number Not Provisioned during validation or user assignment.

Each calling location must have an emergency address before users at that location are fully enabled for outbound calling. Missing or incomplete emergency address data can trigger MTG-CALL-7010 Emergency Address Missing.

The outbound prefix "9" is optional and is configured per dial plan. NetCore Calling 5.0 does not require the prefix globally, and mixed locations may use different prefix settings.

Location-based dial plans apply to users through their assigned location. If a user is moved to another location, revalidate the user number, extension, emergency address inheritance, and dial plan mapping.

This article covers number provisioning and dial plans only. For call forwarding always, busy, or no-answer after 20 s, voicemail, and hunt groups using circular, top-down, or simultaneous routing, see KB-MTG-2005. For the full error-code reference, see KB-MTG-2008.

- Do not assign the same E.164 number to more than one user or service.
- Do not publish a location without a valid emergency address.
- Do not assume prefix "9" is enabled unless the dial plan explicitly shows outboundPrefix: 9.
- Do not delete a number range while any number in the range is assigned.
- Use validation before every publish operation to detect MTG-CALL-7002 Number Not Provisioned and MTG-CALL-7010 Emergency Address Missing.

Common validation failures

Code	Meaning	Typical Cause	Corrective Action
MTG-CALL-7002 Number Not Provisioned	The selected number is not available in the tenant inventory.	Number was typed incorrectly, not imported, already assigned, or not associated with the target location.	Import the E.164 number, correct the assignment, or choose an available number.
MTG-CALL-7010 Emergency Address Missing	The user or location does not have a valid emergency address.	Emergency address was not configured, is incomplete, or was removed during location changes.	Set a complete emergency address on the location and revalidate the configuration.

Related Articles

KB-MTG-2005: Call Forwarding, Hunt Groups and Voicemail Setup

KB-MTG-2008: NetCore Meet and Calling Error Code Reference